TEMA 1 ~ Analiza algoritmilor

Exercitiul 1

A[- matrice m×m
  - nr intregi

a) Algoritm de afișare a tuturor numerelor pare până la întâlnirea primului număr negativ.

```
void alg (int A[m][m], int m){
    int i=0;
    int ok=0;
    while(i<m && ok==0){
        int j=0
        while(j<m && ok==0){
            if(A[i][j]<0)
                ok==1;
            if (A[i][j]%2==0 && ok==0)
                printf("%d",A[i][j]);
            j++;
        }
    i++;
    }
}
```

Constanta	Repetiții
c_1	1
c_2	1
c_3	$p+2$
c_4	$p+1$
c_5	$\sum_{i=0}^{p}(q+2)$
c_6	$\sum_{i=0}^{p}(q+1)$
c_7	$x \to$ boolean
c_8	$\sum_{i=0}^{p}(q+1)$
c_9	$y \to$ boolean
c_{10}	$\sum_{i=0}^{p}(q+1)$
c_{11}	$p+1$

b) TOATE OPERAȚIILE

$$T(m)=f_1+f_2+f_3(p+2)+f_4(p+1)+f_5\cdot\sum_{i=0}^{p}(q+2)+f_6\cdot\sum_{i=0}^{p}(q+1)+f_7\cdot x+f_8\cdot\sum_{i=0}^{p}(q+1)+f_9\cdot y+$$
$$+f_{10}\cdot\sum_{i=0}^{p}(q+1)+f_{11}(p+1)$$

CAZUL CEL MAI FAVORABIL: $p=0, q=0, x=1, y=0$

$$T(m)=f_1+f_2+2f_3+f_4+2f_5+f_6+f_7+f_8+f_{10}+f_{11} \Rightarrow T(m)=\theta(1)$$

CAZUL MEDIU: $p=\frac{m}{2}, q=\frac{m}{2}, x=1, y=\frac{m^2}{4}$

$$T(m)=f_1+f_2+f_3\left(\frac{m}{2}+2\right)+f_4\left(\frac{m}{2}+1\right)+f_5\left(\frac{m}{2}+1\right)^2+f_6\left(\frac{m}{2}+1\right)^2+f_7+f_8\left(\frac{m}{2}+1\right)^2+f_9\cdot\frac{m^2}{4}+$$
$$+f_{10}\left(\frac{m}{2}+1\right)^2+f_{11}\cdot\left(\frac{m}{2}+1\right)$$
$$=f_1+f_2+2f_3+f_4+f_5+f_6+c_7+f_8+f_{10}+f_{11}+\frac{m}{2}(f_3+f_4+2f_5+2f_6+2f_8+f_{10}+f_{11})+$$
$$+\frac{m^2}{4}(f_5+f_6+f_8+f_9+f_{10}) \Rightarrow T(m)=\theta\left(\frac{m^2}{4}\right)=\theta(m^2)$$

CAZUL CEL MAI DEFAVORABIL: $p=n-1$, $q=n-1$, $x=0$, $y=n^2$

$T(n) = f_1 + f_2 + f_3(n+1) + f_4 n + f_5 n(n+1) + f_6 n^2 + f_8 n^2 + f_9 n^2 + f_{10} \cdot n^2 + f_{11} \cdot n =$

$= f_1 + f_2 + f_3 + n(f_3 + f_4 + f_5 + f_{11}) + n^2(f_5 + f_6 + f_8 + f_9 + f_{10})$

$\Rightarrow T(n) = \theta(n^2)$

OPERAȚII CRITICE

CAZUL CEL MAI FAVORABIL: $p=0$, $q=0$, $x=1$, $y=0$

Operațiile critice: $f_1, f_2, f_4, c_6, f_7, c_8, f_{10}, f_{11}$

$T(n) = f_1 + f_2 + f_4 + f_6 + c_7 + f_8 + f_{10} + f_{11} = \theta(1)$

CAZUL MEDIU: $p = \frac{n}{2}$, $q = \frac{n}{2}$, $x=1$, $y = \frac{n^2}{4}$

Operațiile critice: $f_4, f_6, c_8, f_9, f_{10}, f_{11}$

$T(n) = f_4\left(\frac{n}{2}+1\right) + f_6\left(\frac{n}{2}+1\right)^2 + f_8\left(\frac{n}{2}+1\right)^2 + f_9 \cdot \frac{n^2}{4} + f_{10} \cdot \left(\frac{n}{2}+1\right)^2 + f_{11}\left(\frac{n}{2}+1\right)$

$= f_4 + f_6 + f_8 + f_{10} + f_{11} + \frac{n}{2}(f_4 + 2f_6 + 2f_8 + 2f_{10} + f_{11}) + \frac{n^2}{4}(f_6 + f_8 + f_9 + f_{10})$

$\Rightarrow T(n) = \theta\left(\frac{n^2}{4}\right) = \theta(n^2)$

CAZUL CEL MAI DEFAVORABIL: $p=n-1$, $q=n-1$, $x=0$, $y=n^2$

Operațiile critice: $f_4, f_6, f_8, f_9, f_{10}, f_{11}$

$T(n) = f_4 \cdot n + f_6 \cdot n^2 + f_8 n^2 + f_9 n^2 + f_{10} n^2 + f_{11} \cdot n$

$= n(f_4 + f_{11}) + n^2(f_6 + f_8 + f_9 + f_{10}) \quad \Rightarrow T(n) = \theta(n^2)$

Exercitiu 2

$\theta(2n^3+n^2)=\theta(n^3)$

Demonstratie prin dublă incluziune.

Pas I: Aratăm că $\theta(2n^3+n^2)=n^3$

Căutăm $c_1, c_2 \in \mathbb{R}_+^*$, $n_0 \in \mathbb{N}^*$ a.î. $c_1(2n^3+n^2) \leq n^3 \leq c_2(2n^3+n^2)$ $\forall n \geq n_0$

Demonstrăm că pentru $\begin{cases} c_1=0 \\ c_2=1/2 \\ n_0=1 \end{cases}$, inegalitatea este adevărată.

$0 \cdot (2n^3+n^2) \leq n^3 \leq 1/2(2n^3+n^2)$, $\forall n \geq 1$

$0 \leq n^3 \leq n^3 + 1/2 n^2$, $\forall n \geq 1$

$0 \leq 1/2 n^2$, $\forall n \geq 1$

$0 \leq n^2$, $\forall n \geq 1$, adevărat $\Rightarrow \theta(2n^3+n^2)=n^3$

Pas II: Arătăm că $\theta(n^3)=2n^3+n^2$

Căutăm $c_1, c_2 \in \mathbb{R}_+^*$, $n_0 \in \mathbb{N}^*$ a.î. $c_1 n^3 \leq 2n^3+n^2 \leq c_2 n^3$, $\forall n \geq n_0$

Demonstrăm că pentru $\begin{cases} c_1=2 \\ c_2=3 \\ n_0=1 \end{cases}$, inegalitatea este adevărată

$\underbrace{2n^3 \leq 2n^3+n^2}_{I} \leq \underbrace{3n^3}_{II}$, $\forall n \geq 1$

I. $2n^3 \leq 2n^3+n^2$, $\forall n \geq 1$

$0 \leq n^2$, $\forall n \geq 1$, adevărat

II. $2n^3+n^2 \leq 3n^3$, $\forall n \geq 1$

$n^2 \leq n^3$, $\forall n \geq 1$

$1 \leq n$, $\forall n \geq 1$, adevărat

Din I și II $\Rightarrow \theta(n3)=2n^3+n^2$

Pas III: Fie $f(n)=\theta(n^3)$, $n^3=\theta(2n^3+n^2)$ $\Big/$ $\overset{\text{tranzitivitate}}{\Rightarrow}$ $f(n)=\theta(2n^3+n^2) \Rightarrow \theta(n^3) \leq \theta(2n^3+n^2)$ ①

Fie $f(n)=\theta(2n^3+n^2)$, $2n^3+n^2=\theta(n^3)$ $\Big/$ $\overset{\text{tranzitivitate}}{\Rightarrow}$ $f(n)=\theta(n^3) \Rightarrow \theta(2n^3+n^2) \leq \theta(n^3)$ ②

Din ① și ② $\Rightarrow \theta(2n^3+n^2)=\theta(n^3)$

Exercitiu 3

a) $T(m) = \begin{cases} K_1, & m=1 \\ 4 \cdot T(\lfloor m/4 \rfloor) + K_2 m, & m>1 \end{cases} \Rightarrow \begin{cases} T(1) = \theta(1) \\ T(m) = 4T(m/4) + \theta(m) \end{cases}$

METODA ITERATIVĂ

$T(m) = 4T(m/4) + \theta(m) \quad /\cdot 4^0$

$T(m/4) = 4T(m/4^2) + \theta(m/4) \quad /\cdot 4^1$

$T(m/4^2) = 4T(m/4^3) + \theta(m/4^2) \quad /\cdot 4^2$

$\vdots$

$T(m/4^K) = 4T(m/4^{K+1}) + \theta(m/4^K) \quad /\cdot 4^K$

$m = 4^{K+1}$

$K+1 = \frac{1}{2} \lg m \Rightarrow K = \frac{1}{2} \lg m - 1$

$T(m) = 4^{K+1} \cdot T(m/4^{K+1}) + \sum_{i=0}^{K} 4^i \cdot \theta(m/4^i) =$

$= 4^{K+1} \cdot T(1) + \sum_{i=0}^{K} 4^i \cdot \theta(m/4^i) =$

$= 4^{K+1} \cdot T(1) + K \cdot \theta(m) =$

$= 4^{K+1} \cdot \theta(1) + K\theta(m) =$

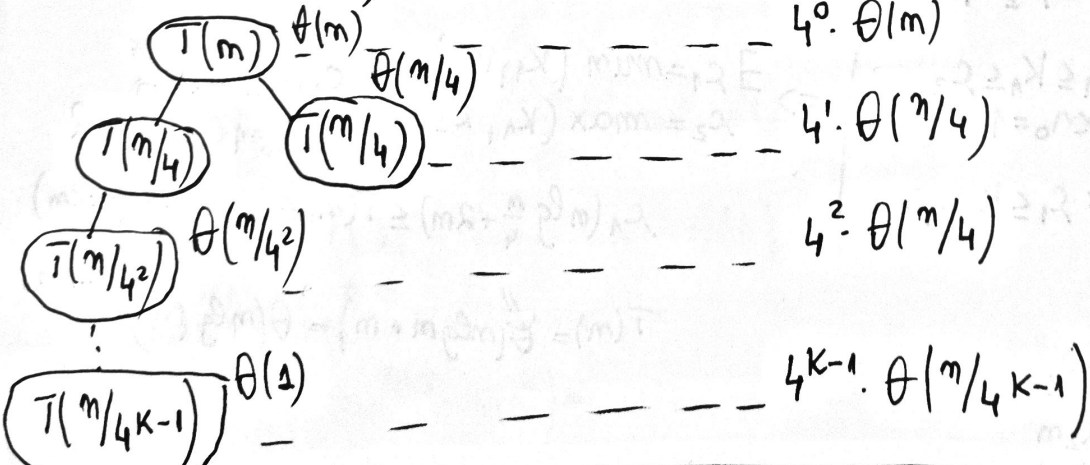
$= m \cdot \theta(1) + \left(\frac{1}{2} \lg m - 1\right) \theta(m) =$

$= m \cdot \theta(1) + \left(\frac{1}{2} \lg m - 1\right) \cdot m \cdot \theta(1) =$

$= \theta(1) \cdot m \left(1 + \frac{1}{2} \lg m - 1\right) = \theta(1) \cdot m \cdot \frac{1}{2} \lg m = \theta\left(\frac{1}{2} m \lg m\right) = \theta(m \lg m)$

METODA ARBORELUI DE RECURENȚĂ

$T(m) = 4T(m/4) + \theta(m), \; T(1) = \theta(1)$



$T(m) = \sum_{i=0}^{K-1} 4^i \cdot \theta(m/4^i) = \sum_{i=0}^{K-1} \frac{4^i}{4^i} \theta(m) = (K-1)\theta(m) = \left(\frac{1}{2} \lg m - 1\right) \theta(m) = \frac{1}{2} \lg m \, \theta(m) =$

$= \theta\left(\frac{1}{2} \lg m \cdot m\right) = \theta(m \lg(m))$

METODA SUBSTITUȚIEI

$$T(n)=\begin{cases} K_1, & n=1 \\ 4T(\lfloor n/4 \rfloor)+K_2 n, & n>1 \end{cases} \Rightarrow \begin{matrix} T(n)=4T(n/4)+\Theta(n) \\ T(1)=K_1 \end{matrix}$$

Alegem $T(n)=\Theta(n\lg n+n)$, $\exists c_1,c_2\in\mathbb{R}_+^*$, $\exists n_0\in\mathbb{N}^*$ a.î.

$$c_1(n\lg n+n)\le T(n)\le c_2(n\lg n+n),\ \forall n\ge n_0$$

Demonstrăm prin inducție inegalitățile de mai sus.

Caz de bază: $n=1$

$$c_1(1\cdot\lg 1+1)\le T(1)\le c_2(1\cdot\lg 1+1)$$

$$c_1\le K_1\le c_2 \Rightarrow n_0=1.$$

Pas de inducție: $n/4\to n$

Ipoteza de inducție: $c_1\left(\frac{n}{4}\lg\frac{n}{4}+\frac{n}{4}\right)\le T\left(\frac{n}{4}\right)\le c_2\left(\frac{n}{4}\lg\frac{n}{4}+\frac{n}{4}\right)$

Arătăm că $c_1(n\lg n+n)\le T(n)\le c_2(n\lg n+n)$

$$c_1\left(\frac{n}{4}\lg\frac{n}{4}+\frac{n}{4}\right)\le T\left(\frac{n}{4}\right)\le c_2\left(\frac{n}{4}\cdot\lg\frac{n}{4}+\frac{n}{4}\right) \quad /\cdot 4 + K_2 n$$

$$c_1\left(n\cdot\lg\frac{n}{4}+n\right)+K_2 n\le T(n)\le c_2\left(n\cdot\lg\frac{n}{4}+n\right)+K_2 n$$

$$c_1\left(n\lg\frac{n}{4}+2n\right)+(K_2-c_1)n\le T(n)\le c_2\left(n\lg\frac{n}{4}+2n\right)+(K_2-c_2)n$$

$$c_1\left(n\lg\frac{n}{4}+2n\right)\le c_1\left(n\lg\frac{n}{4}+2n\right)+\underbrace{(K_2-c_1)}_{\ge 0}n\le T(n)\le c_2\left(n\lg\frac{n}{4}+2n\right)+\underbrace{(K_2-c_2)}_{\le 0}n\le c_2\left(n\lg\frac{n}{4}+2n\right)$$

$$\begin{matrix} K_2-c_1\ge 0 \Rightarrow K_2\ge c_1 \\ K_2-c_2\le 0 \Rightarrow K_2\le c_2 \end{matrix} \Big/ \Rightarrow c_1\le K_2\le c_2$$

Caz de bază: $c_1\le K_1\le c_2$, $n_0=1$

Pas de inducție: $c_1\le K_2\le c_2$

$$\Rightarrow \exists c_1=\min(K_1,K_2),\ c_2=\max(K_1,K_2),\quad c_1,c_2\in\mathbb{R}_+^*,\ \exists n_0=1\in\mathbb{N}^* \text{ a.î.}$$

$$c_1(n\lg\frac{n}{4}+2n)\le T(n)\le c_2(n\lg\frac{n}{4}+2n),\ \forall n\ge n_0$$

$$T(n)=\Theta(n\lg n+n)=\Theta(n\lg n)$$

METODA MASTER

$$T(n)=4T(n/4)+K_2 n$$

$$T(n)=aT(n/b)+f(n)$$

$$\begin{matrix} f(n)=\Theta(n) \\ a=4 \\ b=4 \end{matrix} \Big/ \Rightarrow n^{\log_b a}=n^{\log_4 4}=n^1=n \Rightarrow f(n)=\Theta(n)=n \Rightarrow \text{CAZ 2}$$

Exercitiu 3

b) $T(n) = \begin{cases} K_1, & n=1 \\ 4T(\lceil n/2 \rceil) + T(\lfloor n/2 \rfloor) + K_2 n^2, & n>1 \end{cases} \Rightarrow \begin{cases} T(1) = \Theta(1) \\ T(n) = 5T(n/2) + \Theta(n^2) \end{cases}$

METODA ITERATIVĂ

$T(n) = 5T(n/2) + \Theta(n^2) \quad / \cdot 5^0$

$T(n/2) = 5T(n/2^2) + \Theta(n^2/2^2) \quad / \cdot 5^1$

$T(n/2^2) = 5T(n/2^3) + \Theta(n^2/2^4) \quad / \cdot 5^2$

$\vdots$

$T(n/2^k) = 5T(\overbrace{n/2^{k+1}}^{1}) + \Theta(n^2/2^{2k}) \quad / \cdot 5^k$

$n = 2^{k+1} \Rightarrow k+1 = \lg n$

$k = \lg n - 1$

$$T(n) = 5^{k+1} \cdot T(n/2^{k+1}) + \sum_{i=0}^{k} 5^i \cdot \Theta(n^2/2^{2i}) =$$

$$= 5^{k+1} \cdot T(1) + \sum_{i=0}^{k} \frac{5^i}{4^i} \Theta(n^2) =$$

$$= 5^{k+1} \cdot T(1) + \frac{(\frac{5}{4})^{k+1} - 1}{\frac{5}{4} - 1} \Theta(n^2) =$$

$$= 5^{\lg n} \cdot \Theta(1) + \frac{5^{k+1} - 4^{k+1}}{\frac{1}{4} \cdot 4^{k+1}} \Theta(n^2) =$$

$$= 5^{\lg n} \cdot \Theta(1) + \frac{5^{\lg n} - 4^{\lg n}}{\frac{1}{4} \cdot 4^{\lg n}} \Theta(n^2) =$$

$$= 5^{\lg n} \cdot \Theta(1) + \frac{5^{\lg n} - n^{\lg 4}}{\frac{1}{4} \cdot n^{\lg 4}} \Theta(n^2) =$$

$$= 5^{\lg n} \cdot \Theta(1) + \frac{5^{\lg n} - n^2}{\frac{1}{4} \cdot n^2} \cdot \Theta(n^2) =$$

$$= n^{\lg 5} \cdot \Theta(1) + \frac{5^{\lg n} - n^2}{\frac{1}{4} \cdot n^2} \cdot n^2 \cdot \Theta(1) =$$

$$= n^{\lg 5} \cdot \Theta(1) + 4(n^{\lg 5} - n^2) \cdot \Theta(1) =$$

$$= \Theta(1)(n^{\lg 5} + 4n^{\lg 5} - 4n^2) =$$

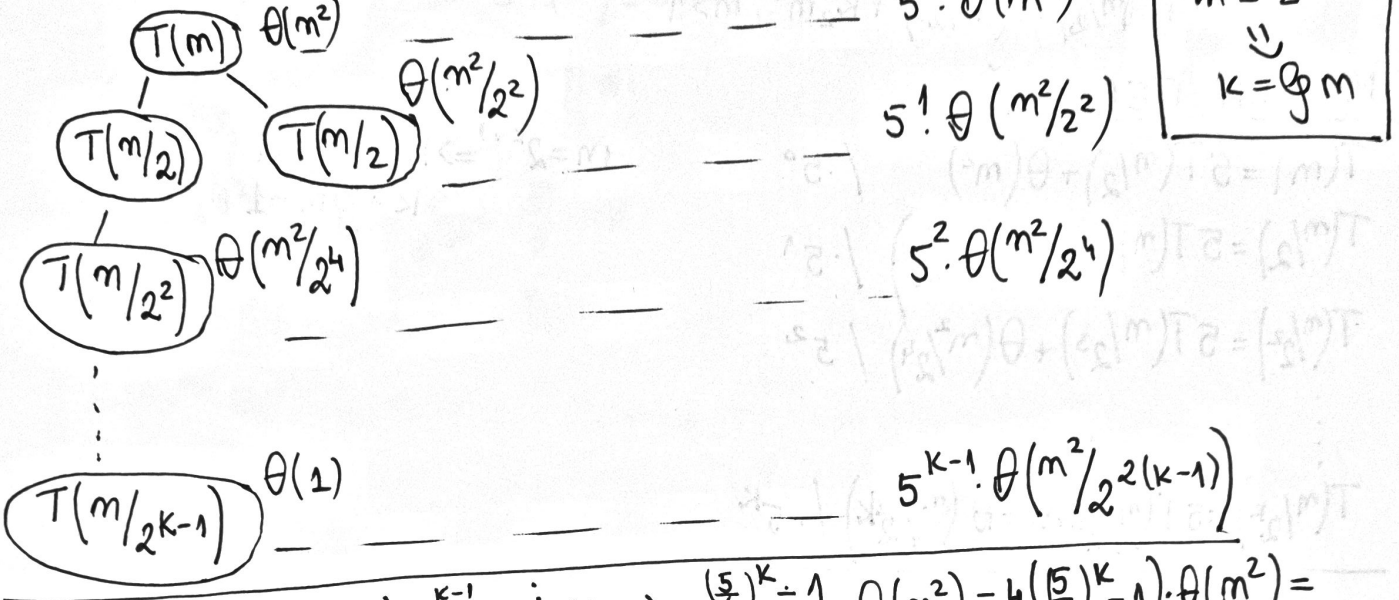
$$= \Theta(1)(5n^{\lg 5} - 4n^2) =$$

$$= \Theta(5n^{\lg 5} - 4n^2) =$$

$$= \Theta(n^{\lg 5})$$

METODA ARBORELUI DE RECURENȚĂ

$T(n) = 5T(n/2) + \Theta(n^2)$, $T(1) = \Theta(1)$



$n = 2^K \Rightarrow K = \log n$

$$T(n) = \sum_{i=0}^{K-1} 5^i \cdot \Theta\left(n^2/2^{2i}\right) = \sum_{i=0}^{K-1} \left(\frac{5}{4}\right)^i \cdot \Theta(n^2) = \frac{\left(\frac{5}{4}\right)^K - 1}{\frac{5}{4} - 1} \Theta(n^2) = 4\left(\left(\frac{5}{4}\right)^K - 1\right) \cdot \Theta(n^2) =$$

$$= 4\left(\frac{5^K - 4^K}{4^K}\right) \cdot \Theta(n^2) = 4 \cdot \frac{5^{\log n} - 4^{\log n}}{4^{\log n}} \cdot n^2 \cdot \Theta(1) =$$

$$= 4 \cdot \frac{n^{\log 5} - n^2}{n^2} \cdot n^2 \cdot \Theta(1) = 4\left(n^{\log 5} - n^2\right) \cdot \Theta(1) = \Theta\left(4n^{\log 5} - 4n^2\right) = \Theta\left(n^{\log 5}\right)$$

METODA SUBSTITUȚIEI

$$T(n) = \begin{cases} K_1, & n = 1 \\ 4T(\lceil n/2 \rceil) + T(\lfloor n/2 \rfloor) + K_2 n^2, & n > 1 \end{cases} \Rightarrow \begin{cases} T(n) = 5T(n/2) + K_2 n^2 \\ T(1) = \Theta(1) \end{cases}$$

Alegem: $T(n) = \Theta(5n^{\log 5} - 4n^2)$, $\exists c_1, c_2 \in \mathbb{R}_+^*$, $\exists n_0 \in \mathbb{N}^*$ a.î.

$$c_1(5n^{\log 5} - 4n^2) \le T(n) \le c_2(5n^{\log 5} - 4n^2), \forall n \ge n_0$$

Demonstrăm prin inducție inegalitățile de mai sus.

Caz de bază: $n = 1$

$$c_1(5 \cdot 1^{\log 5} - 4 \cdot 1^2) \le T(1) \le c_2(5 \cdot 1^{\log 5} - 4 \cdot 1^2)$$

$$c_1(5-4) \le T(1) \le c_2(5-4)$$

$$c_1 \le K_1 \le c_2$$

Pas de inducție: $n/2 \to n$

Ipoteza de inducție: $c_1\left(5 \cdot \left(\frac{n}{2}\right)^{\log 5} - 4 \cdot \frac{n^2}{4}\right) \le T\left(\frac{n}{2}\right) \le c_2\left(5 \cdot \left(\frac{n}{2}\right)^{\log 5} - 4 \cdot \frac{n^2}{4}\right)$

Arătăm că: $c_1(5 \cdot n^{\log 5} - 4n^2) \le T(n) \le c_2(5n^{\log 5} - 4n^2)$

$$c_1\left(5\left(\frac{n}{2}\right)^{\log 5} - 4 \cdot \frac{n^2}{4}\right) \le T\left(\frac{n}{2}\right) \le c_2\left(5\left(\frac{n}{2}\right)^{\log 5} - 4 \cdot \frac{n^2}{4}\right)$$

$$c_1\left(5\cdot\left(\frac{n}{2}\right)^{\lg 5}-n^2\right)\le T\left(\frac{n}{2}\right)\le c_2\left(5\cdot\left(\frac{n}{2}\right)^{\lg 5}-n^2\right)\ /\cdot 5\ /+k_2n^2$$

$$c_1\left(25\cdot\left(\frac{n}{2}\right)^{\lg 5}-5n^2\right)+k_2n^2\le T(n)\le c_2\left(25\cdot\left(\frac{n}{2}\right)^{\lg 5}-5n^2\right)+k_2n^2$$

$$c_1\left(\frac{25}{2^{\lg 5}}\cdot n^{\lg 5}-5n^2\right)+k_2n^2\le T(n)\le c_2\left(\frac{25}{2^{\lg 5}}n^{\lg 5}-5n^2\right)+k_2n^2$$

$$2^{\lg 5}=5^{\lg 2}=5 \Rightarrow \frac{25}{2^{\lg 5}}=\frac{25}{5}=5$$

$$c_1\left(5n^{\lg 5}-5n^2\right)+k_2n^2\le T(n)\le c_2\left(5n^{\lg 5}-5n^2\right)+k_2n^2$$

$$c_1\left(5n^{\lg 5}-6n^2\right)+(k_2-c_1)n^2\le T(n)\le c_2\left(5n^{\lg 5}-6n^2\right)+(k_2-c_2)n^2$$

$$c_1\left(5n^{\lg 5}-6n^2\right)\le c_1\left(5n^{\lg 5}-6n^2\right)+\underbrace{(k_2-c_1)}_{\ge 0}n^2\le T(n)\le c_2\left(5n^{\lg 5}-6n^2\right)+\underbrace{(k_2-c_2)}_{\le 0}n^2\le c_2\left(5n^{\lg 5}-6n^2\right)$$

$k_2-c_1\ge 0 \Rightarrow k_2\ge c_1$
$k_2-c_2\le 0 \Rightarrow k_2\le c_2$ $\Big/ \Rightarrow c_1\le k_2\le c_2$

Caz de bază: $c_1\le k_1\le k_2$
Pas de inducție: $c_1\le k_2\le c_2$ $\Big| \Rightarrow \exists\, c_1=\min(k_1,k_2)$, $c_2=\max(k_1,k_2)$ $\quad c_1,c_2\in\mathbb{R}_+^*$

$\exists\, n_0=1\in\mathbb{N}^*$ a.î.

$$c_1\left(5n^{\lg 5}-6n^2\right)\le T(n)\le c_2\left(5n^{\lg 5}-6n^2\right)\quad \forall n\ge n_0$$

$$T(n)=\Theta\left(5n^{\lg 5}-6n^2\right)=\Theta\left(n^{\lg 5}\right)$$

METODA MASTER

$T(n)=5T(n/2)+k_2n^2$
$T(n)=aT(n/b)+f(n^d)$

Caz 2: Dacă $f(n)=\Theta\left(n^{\log_b a}\right)$ atunci $T(n)=\Theta\left(n^{\log_b a}\cdot\lg n\right)$

$$\begin{cases} f(n)=\Theta(n^2) \\ a=5 \\ b=2 \end{cases} \Rightarrow n^{\log_b a}=n^{\log_2 5}=n^{\lg 5}$$

Cum $n^{\lg 5}\ne\Theta(n^2)$ $\Rightarrow$ nu putem aplica cazul 2 Master $\Rightarrow$ aplicăm cazul 1

$T(n)=aT(n/b)+f(n)$
Caz 1: Dacă $\exists\,\varepsilon>0$ a.î. $f(n)=O\left(n^{\log_b a-\varepsilon}\right)$ atunci $T(n)=\Theta\left(n^{\log_b a}\right)$

$$\begin{cases} f(n)=\Theta(n^2) \\ a=5 \\ b=2 \end{cases} \Rightarrow n^{\log_b a}=n^{\log_2 5}=n^{\lg 5}$$

$f(m) = \theta(m^2)$

Căutăm $\varepsilon > 0$ a.î. $2 = \log_2 5 - \varepsilon$.

$2 = \lg 5 - \varepsilon \Leftrightarrow \varepsilon = \lg 5 - 2 \Leftrightarrow \varepsilon = \lg 5 - \lg 4 = \lg \frac{5}{4} > \lg 1 = 0 \Rightarrow \varepsilon > 0$

$f(m) = \theta(m^2) = \theta\left(m^{\lg 5 - \lg \frac{5}{4}}\right)$

$f(m) = \theta\left(m^{\lg 5 - \lg \frac{5}{4}}\right) \Rightarrow f(m) = O\left(m^{\lg 5 - \lg \frac{5}{4}}\right) \Rightarrow \exists\, \varepsilon = \lg \frac{5}{4}$ a.î. $f(m) = O\left(m^{\log_b a} - m^{\log_2 \frac{5}{4}}\right)$

↳ caz 1 Master

$T(m) = \theta(m^{\log_2 \frac{5}{4}}) = \theta(m^{\lg 5})$